

Emilio Daza

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Dartmouth computer science student graduating June 2027 with experience building cross-platform applications, backend systems, intelligent payment-routing models, and healthcare ML pipelines for heart-disease prediction and brain-tumor MRI classification.

Education

Dartmouth College

Hanover, NH

B.A. in Computer Science, Minor in Statistics

Expected Jun. 2027

Relevant Coursework: Computer Architecture, Compilers, Discrete Mathematics, Robotics Design & Programming, Computer Vision, Multivariate Statistics & Statistical Learning, and Probability & Statistical Inference.

Beca Cometa Scholar (2023): 1 of 2 students selected nationwide in Peru for a full undergraduate scholarship at a top U.S. university.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, Java, C, C++, Dart, R

Frameworks: React Native, Expo, Flutter, React, PyTorch, Gymnasium, R-Studio

Backend/Data: ClickHouse, PostgreSQL, Supabase, Firebase, REST APIs, Edge Functions, row-level security

Tools/Systems: AWS, Docker, Linux, Git, GitHub, NetworkX, ROS 2, Gazebo

Experience

Izipay

Lima, Peru

Machine Learning Engineer Intern

Jun. 2026 – Aug. 2026

- Developed and evaluated an RL-based intelligent payment-routing prototype in a synthetic simulator for Izipay, a leading Peruvian processor serving 1.2M+ merchants; now adapting the architecture by querying transaction-node data from an AWS-hosted ClickHouse instance using SQL.
- Modeled processor selection under outages, latency variation, technical failures, and payment rejection as a Markov decision process; developed a custom Gymnasium environment and PyTorch Double DQN agent over a NetworkX topology with action masking, experience replay, and target networks.
- Evaluated the learned policy across 1,000 seeded episodes generated from synthetic routing probabilities: 92.3% simulated approval and 178.4 ms mean latency; automated JSON/Matplotlib reporting and presented results to senior strategy and business-solutions leaders.

AlgoVerse AI Research Program

Remote

Machine Learning Researcher

Jun. 2026 – Sep. 2026

- Developing a formal research project on whether finite prefixes of ordinary RL training can distinguish reward misspecification from task-relevant observation insufficiency, and what minimum diagnostic interventions make the failure modes identifiable.
- Collaborating with a three-person ML research team under the mentorship of a Google Cloud AI & Infrastructure Software Engineer; selected for a 12-week publication-oriented program with a competitive 30% tuition scholarship.

Dartmouth Center for Technology and Behavioral Health

Hanover, NH

Software Engineer, Cross-Platform App

Nov. 2025 – Mar. 2026

- Implemented and validated Flutter desktop functionality on Arch Linux for Evergreen, a research-backed platform supporting personalized student well-being.
- Designed a dual-mode interface separating internal diagnostic and developer workflows from the participant-facing user experience.

Dartmouth College, E.E. Just Research Program

Hanover, NH

Funded Machine Learning Research Intern

Jun. 2024 – Aug. 2024

- Selected as 1 of 8 first-year students for a funded summer program; built PyTorch feedforward and CNN pipelines for heart-disease prediction and four-class brain-tumor MRI classification, then presented the research at the largest math conference in the world: the Joint Mathematics Meeting in 2025.

Selected Projects

Thyme – Co-Founder & Founding Software Engineer

Apr. 2026 – Present

- Built a TypeScript React Native/Expo application for iOS and Android with design and public-health teammates; distributed an iOS beta through TestFlight and an Android APK for internal testing.
- Designed a Supabase/PostgreSQL backend with authentication, row-level security, Edge Functions, community and menu data, and remote version controls; implemented persistent profiles, settings, and nutrition logs.
- Integrated OpenAI APIs for multimodal meal and menu analysis, bilingual speech transcription and text-to-speech, and generated cooking guidance; developed location-aware dining discovery using Google Places and campus-menu data sources.

Robotics Mapping, Planning, and Perception Systems

Spring 2026

- Implemented ROS 2 systems in Gazebo and Stage for odometry, coordinate transforms, LiDAR processing, and occupancy-grid mapping; a YOLO11n and FastSAM perception pipeline producing target bounding boxes and segmentation masks from camera video streams; BFS, DFS, and A* planners plus a LiDAR mapper using Bresenham raycasting, log-odds updates, and dynamically expanding occupancy grids.